Physics

Scientific Inquiry

Grade 6-7

Quarter 1

Name: _____ Class: _____ Date: _____

Designed by Sunquist in Park Lane Harbor

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Unit 1 — Scientific Inquiry

Scientific Inquiry — Observations

Lesson 1 of 6

Distinguish qualitative from quantitative observations and objective from subjective statements, and explain what makes a statement scientific.

Observations

> Note
> An **observation** is the act of gathering information using your senses or instruments. Information gathered from an observation is given as a **statement**.

Qualitative and Quantitative Statements

> Note
> **Qualitative**: descriptive, non-numerical data.
> **Quantitative**: numerical, measured data.

Qualitative	Quantitative
the solution is blue	the solution has volume $50.0\ \mathrm{mL}$
the object is hot	the temperature is $80^\circ\mathrm{C}$
the surface is rough	the roughness average is $2.3\ \mu\mathrm{m}$
the sound is loud	the sound intensity is $85\ \mathrm{dB}$

Subjective and Objective Statements

> Note
> **Subjective**: opinion-based statements that vary by person. **Objective**: fact-based, verifiable statements.

Subjective	Objective
the soup tastes salty	the soup contains $800\ \mathrm{mg}$ of sodium
the room feels warm	the room temperature is $24^\circ\mathrm{C}$
chocolate is the best flavor	chocolate is the most-sold flavor
Pikachu is cute	Pikachu is yellow

> Warning
> Objective statements can be qualitaitve.

Statement	Qualitative (observable quality)	Objective (verifiable fact)
Pikachu is yellow	describes color	color is measurable by wavelength
Water is wet	describes a feel	wetness is a physical property
Iron is magnetic	describes behavior	magnetism is testable

Can a quantitative statement be subjective?

Answer: No, but it can be uncertain.

What makes a statement *scientific*?

> Note
> A **scientific statement** must be **objective**, **testable**, and **falsifiable**.
> - **Testable**: data can be gathered through observation or experiment.
> - **Falsifiable**: it is *possible* to be proven false through testing.

Examples

Non-scientific	Scientific
The red car is fast.	The red car has a top speed of $210\ \frac{km}{h}$ (quantitative)
The water feels hot.	The water is boiling and producing steam. (qualitative)
This is a heavy backpack.	The backpack has a mass of $12\ \mathrm{kg}$. (quantitative)
The flower is pretty.	The flower has pink petals. (qualitative)

Scientific Inquiry — Measurement & Metric Prefixes

Lesson 2 of 6

Measure with the correct tools, distinguish accuracy from precision, and convert between metric prefixes and scientific notation.

What Is a Measurement?

> Measurement
> An observation that compares an unknown quantity to a known quantity.

Tool	Quantity Measured	SI Units
Meter stick	distance, displacement	meter, m
Stopwatch	time	second, s
Mass scale	mass	kilogram, kg
Force scale	force	Newton, N

Measuring Tools



Accuracy

> **Accuracy**
> How close the measurements are to the actual value.

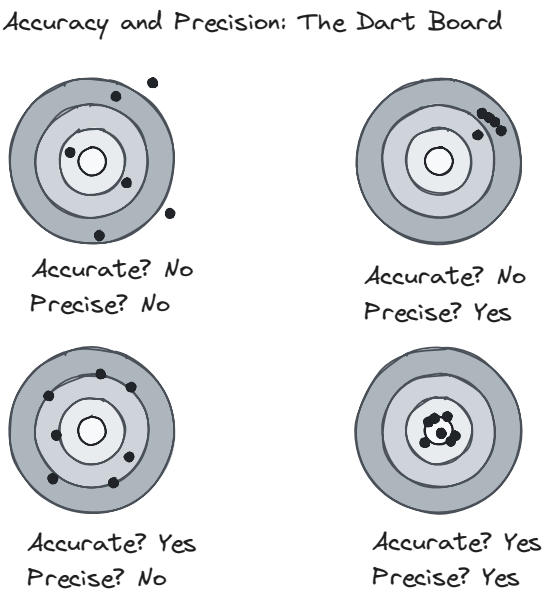
Lower Accuracy (Actual: 30 kg)	Higher Accuracy (Actual: 30 kg)
29 kg	29 kg
30 kg	29 kg
28 kg	30 kg
28 kg	31 kg
Average: 28.8 kg	**Average:** 29.8 kg

Precision

> **Precision**
> How close together the measurements are to each other.

Lower Precision	Higher Precision
50 mm	54 mm
54 mm	56 mm
59 mm	55 mm
56 mm	56 mm
Range: 9 mm	**Range:** 2 mm

Accuracy vs Precision



Metric System

> **Metric System**
> A decimal-based system of measurement used in science — built on powers of 10 and used by scientists worldwide.

- The wavelength is $0.000000673\ \text{m} \rightarrow 673\ \text{nm}$
- My phone's storage is $512\ \text{GB}$: $1\ \text{GB} = 10^9$ bytes .

> Prefixes make very large or very small numbers easier to report.

Metric Prefixes

Prefix	Symbol	Order of Magnitude
Tera-	T	10^{12}
Giga-	G	10^{9}
Mega-	M	10^{6}
Kilo-	**K**	10^{3}
hecto-	**h**	10^{2}
deka-	**da**	10^{1}
(none)	**(base)**	$10^{0} = 1$
deci-	**d**	10^{-1}
centi-	**c**	10^{-2}
milli-	**m**	10^{-3}
micro-	μ	10^{-6}
nano-	n	10^{-9}
pico-	p	10^{-12}

Scientific Notation

> **Scientific Notation**
> Writing numbers in the form of $a \times 10^n$
> The exponent n moves the decimal point.

Number	Scientific Notation
$602{,}000{,}000{,}000{,}000{,}000{,}000$	6.02×10^{23}
0.000000000001	1.0×10^{-12}

> A metric prefix is just a scientific-notation scale.
> Kilo $= 10^3$, nano $= 10^{-9}$

Big Numbers → Scientific Notation

> Move the decimal point left
> The number of places moved equals a positive exponent.

$$9{,}300{,}000 = 9.3 \times 10^6$$

$$150{,}000{,}000 = 1.5 \times 10^8$$

Small Numbers → Scientific Notation

> **Move the decimal point right.**
> The number of places moved equals a negative exponent.

$$0.0000045 = 4.5 \times 10^{-6}$$

$$0.000000025 = 2.5 \times 10^{-8}$$

Big Prefixes → Scientific Notation

> **Big prefixes give a positive power of ten.**
> The measurement becomes $a \times 10^{n}$ where $n > 0$.

$$3.0\ \text{Gm} = 3.0 \times 10^{9}\ \text{m}$$

$$2.5\ \text{Tm} = 2.5 \times 10^{12}\ \text{m}$$

Small Prefixes → Scientific Notation

> **Small prefixes give negative powers of ten.**
> The measurement becomes $a \times 10^{n}$ where $n < 0$.

$$4.2\ \mu\text{s} = 4.2 \times 10^{-6}\ \text{s}$$

$$750\ \text{nm} = 7.5 \times 10^{-7}\ \text{m}$$

Example: Converting Metric Prefixes

5 kilometers to meters.

$$5\ \cancel{\text{km}}\left(\frac{1000\ \text{m}}{1\ \cancel{\text{km}}}\right) = 5000\ \text{m}$$

Example: Converting Metric Prefixes

15 kilograms to milligrams.

$$15\ \cancel{\text{kg}}\left(\frac{1000\ \cancel{\text{g}}}{1\ \cancel{\text{kg}}}\right)\left(\frac{1000\ \text{mg}}{1\ \cancel{\text{g}}}\right) = 15{,}000{,}000\ \text{mg}$$

Example: Converting Time Units

2.5 hours to seconds.

$$2.5\ \cancel{\text{hr}}\left(\frac{60\ \cancel{\text{min}}}{1\ \cancel{\text{hr}}}\right)\left(\frac{60\ \text{s}}{1\ \cancel{\text{min}}}\right) = 9000\ \text{s}$$

Example: Converting Speed Units

40 km/h to m/s.

$40\ \frac{\cancel{\text{km}}}{\cancel{\text{h}}}\left(\frac{1000\ \text{m}}{1\ \cancel{\text{km}}}\right)\left(\frac{1\ \cancel{\text{h}}}{3600\ \text{s}}\right) = \frac{40 \times 1000\ \text{m}}{3600\ \text{s}} = 11.1\ \frac{\text{m}}{\text{s}}$

Scientific Inquiry – Base Units & Derived Units

Lesson 3 of 6

Distinguish base units from derived units and derive the unit of a quantity from its formula.

International System of Units (SI)

> **There are seven SI base units.**

Quantity	Unit	Symbol
length or distance	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

Why Do We Need Other Units?

Imagine describing a car's speed like this:

> "The car travels 100 meters every 5 seconds."

Instead, we can express this with a **derived unit**: **meters per second** $\left(\frac{\text{m}}{\text{s}}\right)$.

> **Derived Unit**
> A combination of base units.

How Derived Units Work

- A derived unit comes from the **formula** that defines the quantity.

Example: Area

1. **Formula:** $\text{Area} = \text{length} \times \text{width}$
2. **Base Units:** Both length and width are measured in m (meters).
3. Area is measured with a **derived unit:** $\text{m} \times \text{m} = \text{m}^2$ (meters squared).

To find a derived unit:

> **Tip**
> 1. **Find the formula** for the quantity.
> 2. **Replace** each quantity with its **SI base unit**.
> 3. **Simplify** the expression.

Example 1: Velocity

- The formula for **average velocity** is:

$$\bar{v} = \frac{\Delta x}{t}$$

$$\bar{v} = \frac{\Delta x}{t}$$

- Δx, displacement, has the unit **meters** (m)
- t, time, has the unit **seconds** (s) .

$$\bar{v} = \frac{\mathrm{m}}{\mathrm{s}}$$

$$\bar{v} = \frac{\mathrm{m}}{\mathrm{s}}$$

- *meters per second*

Example 2: Acceleration

- The formula for **acceleration** is:

$$a = \frac{\Delta v}{t}$$

$$a = \frac{\Delta v}{t}$$

- Δv, change in velocity, has the unit $\frac{\mathrm{m}}{\mathrm{s}}$.
- t has the unit s.

$$a = \frac{\frac{\mathrm{m}}{\mathrm{s}}}{\mathrm{s}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

$$a = \frac{\frac{\mathrm{m}}{\mathrm{s}}}{\mathrm{s}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

- *meters per second squared*

Example 3: Force

- The formula for **force** is:

$$F = m \times a$$

$$F = m \times a$$

- m has the unit kg
- a has the unit $\frac{\mathrm{m}}{\mathrm{s}^2}$.

$$F = \mathrm{kg} \cdot \frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

$$F = \mathrm{kg} \cdot \frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

- *kilograms meters per second squared*, *Newton*

Special Derived Units

> Some derived units appear so often that they get their own names.

Derived Quantity	Formula	Base Units	Special Name & Symbol
Force	$F = ma$	$\mathrm{kg} \cdot \frac{\mathrm{m}}{\mathrm{s}^2}$	**Newton** (N)
Energy/Work	$W = Fd$	$\mathrm{kg} \cdot \frac{\mathrm{m}^2}{\mathrm{s}^2}$	**Joule** (J)
Power	$P = \frac{W}{t}$	$\mathrm{kg} \cdot \frac{\mathrm{m}^2}{\mathrm{s}^3}$	**Watt** (W)
Frequency	$f = \frac{1}{T}$	$\frac{1}{\mathrm{s}}$	**Hertz** (Hz)
Charge	$Q = I \cdot t$	$\mathrm{A} \cdot \mathrm{s}$	**Coulomb** (C)

The Ampere

The **ampere** (A) is one of the seven SI base units, but it is also defined by the coulomb. From charge $Q = I\,t$ (with t in seconds):

$$1\ \mathrm{A} = 1\ \frac{\mathrm{C}}{\mathrm{s}}$$

One ampere is one coulomb of charge passing each second.

Scientific Inquiry – Problem Solving

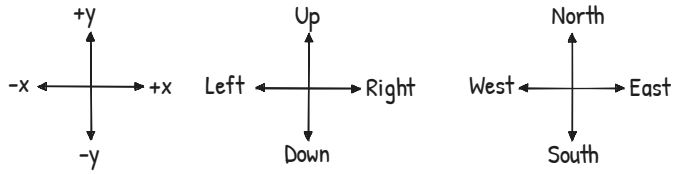
Lesson 4 of 6

Classify quantities as scalars or vectors and explain the steps used to solve a physics problem.

Scalars and Vectors

> Scalar
> A quantity with a magnitude only (e.g. **speed** of $90\frac{\mathrm{km}}{\mathrm{h}}$).

> Vector
> A quantity magnitude *and direction* (e.g. **velocity** of $90\frac{\mathrm{km}}{\mathrm{h}}$ *east*).



List of Scalars and Vectors

Scalar Quantities	Vector Quantities
distance	displacement
speed	velocity
energy/work	acceleration
charge	force
	momentum

Problem Solving Steps

> Important
>
> 1. **Annotate the problem to find the quantities.**
> 2. **What equation connects the quantities?**
> 3. **Plug in with units and solve.**
> 4. **Check the units and direction (if applicable).**

Example 1

A train travels $180\ \text{km}$ in $2\ \text{h}$. What is its speed?

$s = \frac{d}{t} = \frac{180\ \text{km}}{2\ \text{h}} = 90\frac{\text{km}}{\text{h}}$

s: $\frac{\text{km}}{\text{h}}$, scalar

Example 2

A bird flies $500\ \text{m east}$ in $25\ \text{s}$. What is its velocity?

$v = \frac{\Delta x}{t} = \frac{500\ \text{m}}{25\ \text{s}} = 20\frac{\text{m}}{\text{s}}\ \text{east}$

v: $\frac{\text{m}}{\text{s}}$, east

Example 3

A car goes from $10\frac{\text{m}}{\text{s}}$ to $30\frac{\text{m}}{\text{s}}$ in $5\ \text{s}$. What is its acceleration?

$a = \frac{v - v_0}{t} = \frac{30\frac{\text{m}}{\text{s}} - 10\frac{\text{m}}{\text{s}}}{5\ \text{s}} = \frac{20\frac{\text{m}}{\text{s}}}{5\ \text{s}} = +4\frac{\text{m}}{\text{s}^2}$

a: $\frac{\text{m}}{\text{s}^2}$, positive

Scientific Inquiry – Graphs

Lesson 5 of 6

Build and read graphs with SULTAN, find the line of best fit, y-intercept, and slope, and extrapolate from a graph.

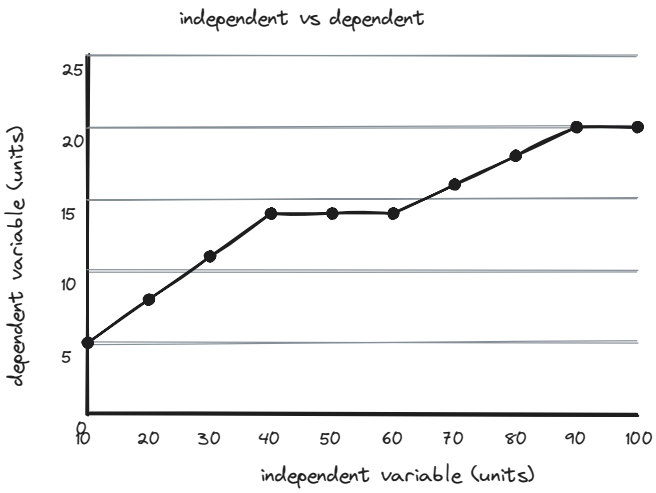
Tables

- The independent variable goes in the first column.
- **average (mean)**: add all the values, then divide by how many there are.

Name of Pet	Species	Color of Eyes	Mass (kg)
Toby	Cat	Yellow	4.2
Sugar	Dog	Brown	15.1
Hiro	Dog	Dark Brown	23.8
Tiger	Cat	Yellow	6.0
Booboo	Cat	Brown	3.5

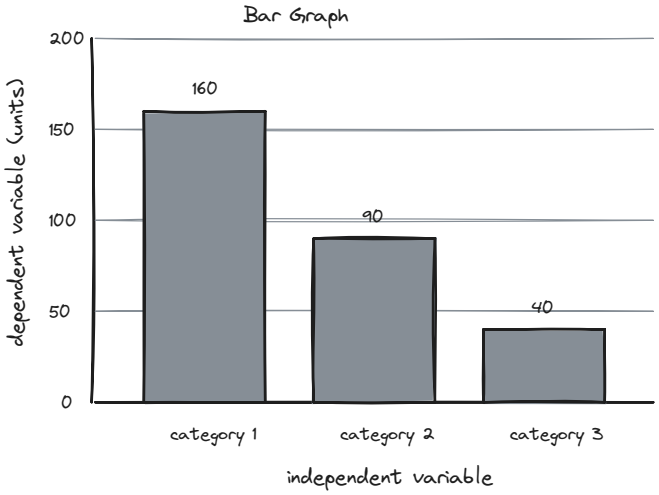
Line Graph

- Used for continuous data.



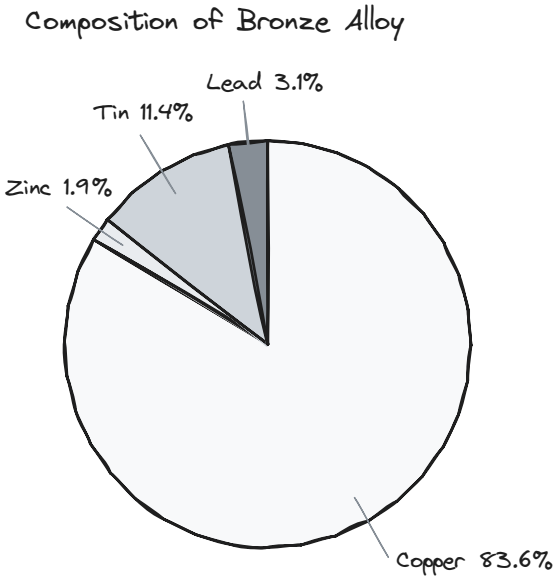
Bar Graph

- Used for categories and comparisons.



Pie Chart

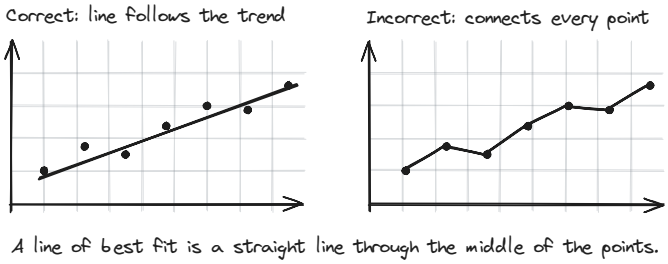
- Shows parts of a whole (percentages).



SULTAN

> **Requirements for a graph:**
- **S**cale: Choose appropriate axis scales
- **U**nits: Label both axes with units
- **L**egend: Include if multiple data sets
- **T**itle: Describe what the graph shows
- **A**xes: Label independent (x) and dependent (y)
- **N**eat: Plot points clearly

Line of Best Fit



> **Warning**
> Do **not** connect the points one by one — that is a *line graph*, not a **line of best fit**. Draw one straight line that follows the trend.

Scientific Inquiry – Scientific Method & Variables

Lesson 6 of 6

Write a hypothesis, identify variables and control and experimental groups, and evaluate the hypothesis after the experiment.

The Scientific Method

> **Scientific Method**
> **Observation → Question → Hypothesis → Experiment → Conclusion.**

Step	What happens
Observation	Spot patterns, trends, or anomalies in data.
Question	Ask a specific, testable question based on observations.
Hypothesis	Offer a testable possible explanation.
Experiment	Run a controlled test, varying one factor at a time and measuring outcomes.
Conclusion	Analyze results to see if they support the hypothesis, contradict it, or prompt new questions.

Hypothesis

> **Note**
> A **hypothesis** is a **testable** and **falsifiable** prediction based on observations — **not** a random guess. Write it as an "**if... then...**" statement (e.g. *If* the ramp is longer, *then* the ball will roll farther).

Variables

- Independent Variable: The variable you change on purpose. Plotted on the x-axis.
- Dependent Variable: The variable you measure. Plotted on the y-axis.
- Control Variable: A variable kept constant.
- Control Group: The group that does not receive the treatment; used as a baseline for comparison.
- Experimental Group: The group that receives the treatment or independent variable.

Why Control Variables?

To run a **fair** and **repeatable** test, keep every other variable the same:

- Only the **independent variable** changes, so a change in the **dependent variable** can be traced to that one change.
- If several things changed at once, you could not tell which one caused the result.
- Another person in a different lab can repeat the test and get the same result.

Example: A student drops a ball from different heights and measures how high it bounces.

- **Question:** Does the drop height affect how high the ball bounces?
- **Hypothesis:** If the ball is dropped from a greater height, then it will bounce higher.
- **Independent variable:** drop height (what we change)
- **Dependent variable:** bounce height (what we measure)
- **Control variable:** the **same ball** (kept the same)
- **Experimental group:** the drop from a **higher** height (gets the change)
- **Control group:** the drop from the **original** height (baseline)

After the Experiment

- Compare data against your hypothesis.
- Was the hypothesis supported or not supported?
- A conclusion states what the evidence showed — even if the hypothesis was wrong.